DO NOW — RECALL

Software development: practical testing

Name: _____ Class: _____ Date: _____

1. What is a prototype?

2. Why might a tester run tests at each sprint?

3. What does AND require for a compound condition to be true?

4. For valid ages 12–16 inclusive, which four values test either side of both boundaries?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An early model that users evaluate and refine.

2. Model answer

To catch regressions while new features are introduced.

3. Model answer

Both conditions must be true.

4. Model answer

11, 12, 16 and 17.

Vocabulary — know these terms

normal test — Typical valid input.

boundary test — Input at or near a valid limit.

erroneous test — Input outside permitted requirements.

expected result — Predicted correct output for a test.

regression — New change breaks previously working behaviour.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and explain

Upper boundary 30 is accepted while 31 is invalid; 16 is a representative normal value.

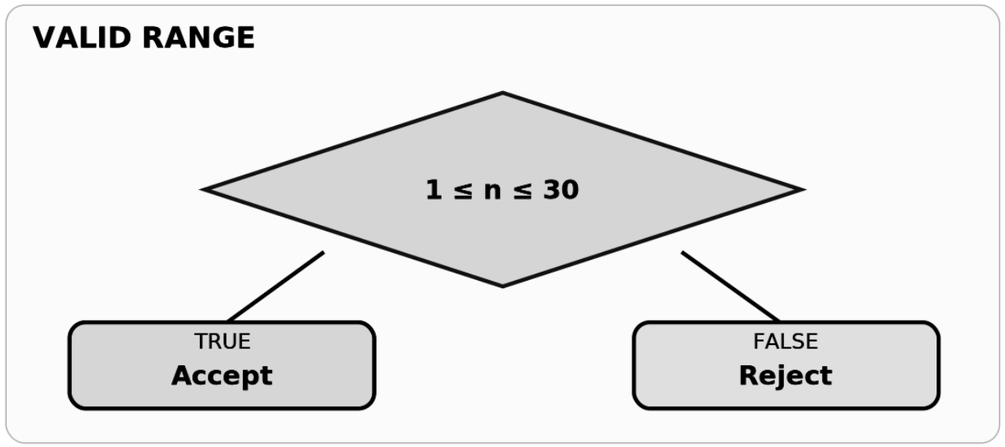
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose boundary and invalid tests for a class size of 1–30 inclusive.

Lower boundary 1 is accepted while 0 is invalid.



VALID RANGE: condition 1 ≤ n ≤ 30 yes Accept no Reject

NEW LEARNING

Read, annotate and explain

State the valid range

The specification permits whole-number marks from 0 through 100 inclusive.

Choose targeted tests

Normal value 50; valid boundaries 0 and 100; invalid just beyond boundaries −1 and 101.

Write expected results

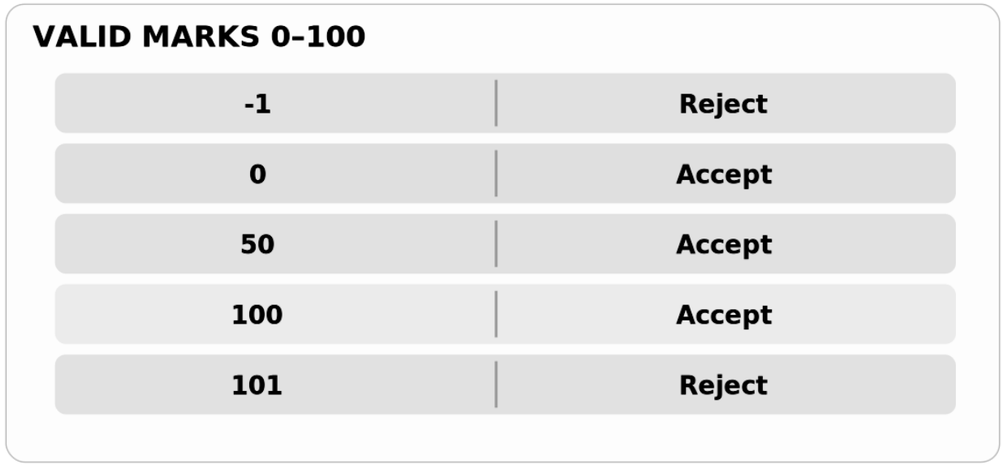
A useful test log records input, expected outcome and actual outcome; a mismatch is evidence of a fault.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

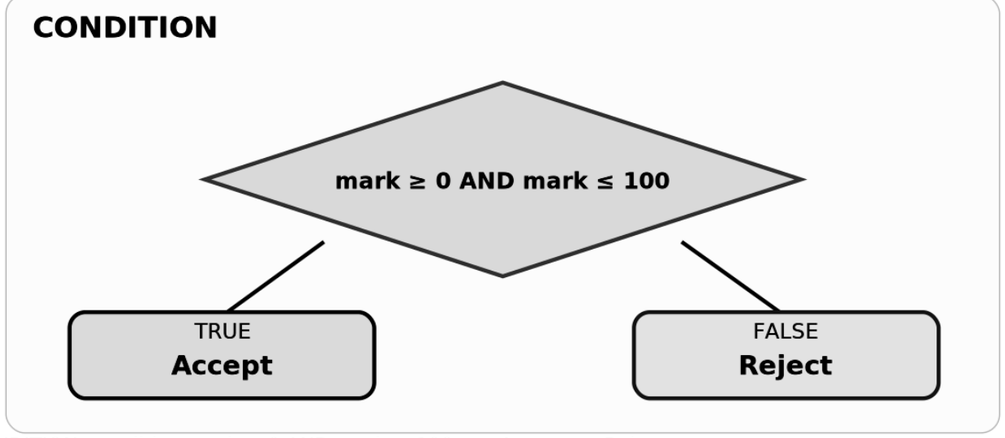
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



VALID MARKS 0–100: -1 Reject; 0 Accept; 50 Accept; 100 Accept; 101 Reject

New Learning



CONDITION: condition mark ≥ 0 AND mark ≤ 100 yes Accept no Reject

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Testing only typical valid inputs proves that validation rejects out-of-range values.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A developer tests values 20,30,40 for a permitted range 20–40. Explain why this is insufficient and propose a stronger suite.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

TEST RECORD

Input	**0**
Expected	**true**
Actual	**true**
Outcome	**Pass**

TEST RECORD: Input 0; Expected true; Actual true; Outcome Pass

I DO — WORKED EXAMPLE

Follow the worked method

Design tests for a password length from 8 through 20 characters inclusive.

Predict first

Test just below and at the lower limit: 7 rejected, 8 accepted.

Model and explain

Also test a normal 14, upper boundary 20, and 21 rejected.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A whole-number temperature must be 10–40 inclusive. Give a normal, a lower-boundary, an upper-boundary and an erroneous test.

Question 2

Write C# to validate a score from 1 to 9 inclusive and give four expected-result test pairs.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Lower boundary 1 is accepted while 0 is invalid.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

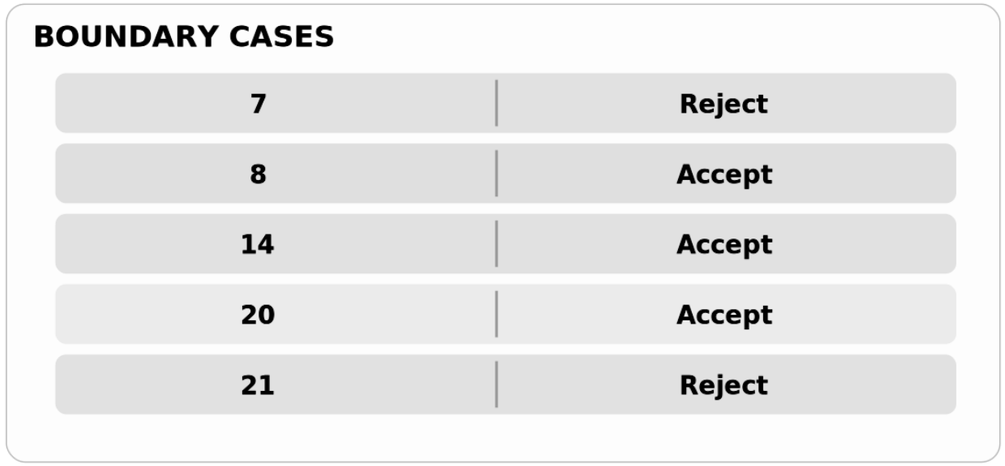
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Design tests for a password length from 8 through 20 characters inclusive.

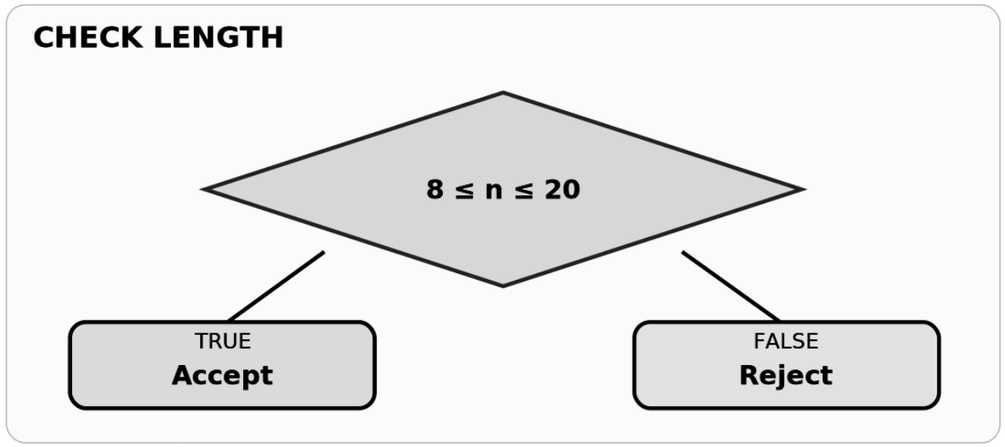
Test just below and at the lower limit: 7 rejected, 8 accepted.



BOUNDARY CASES: 7 Reject; 8 Accept; 14 Accept; 20 Accept; 21 Reject

Design tests for a password length from 8 through 20 characters inclusive.

Test just below and at the lower limit: 7 rejected, 8 accepted.



CHECK LENGTH: condition 8 ≤ n ≤ 20 yes Accept no Reject

WE DO — GUIDED PRACTICE

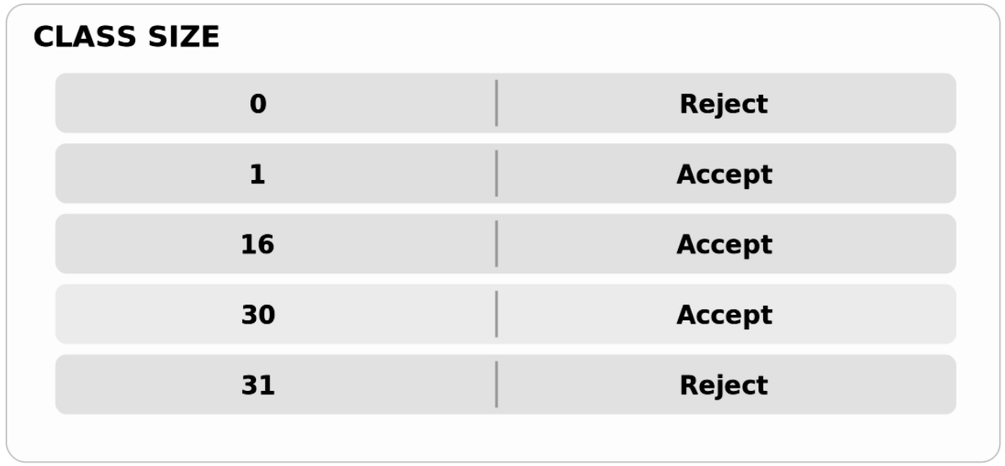
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose boundary and invalid tests for a class size of 1–30 inclusive.

Lower boundary 1 is accepted while 0 is invalid.



CLASS SIZE: 0 Reject; 1 Accept; 16 Accept; 30 Accept; 31 Reject